MEC4126F: Integrated Embedded Systems

Prac 7
08 May 2025

Total marks: 51

Instructions to students

1. This template file contains space for the answers to the written questions of Prac 7.
2. Ensure that you copy-paste your answers inside the space allocated for each question.
3. Provide your numerical answers to **TWO (2)** significant decimal points, unless stated otherwise.

PeopleSoft ID: 1878246

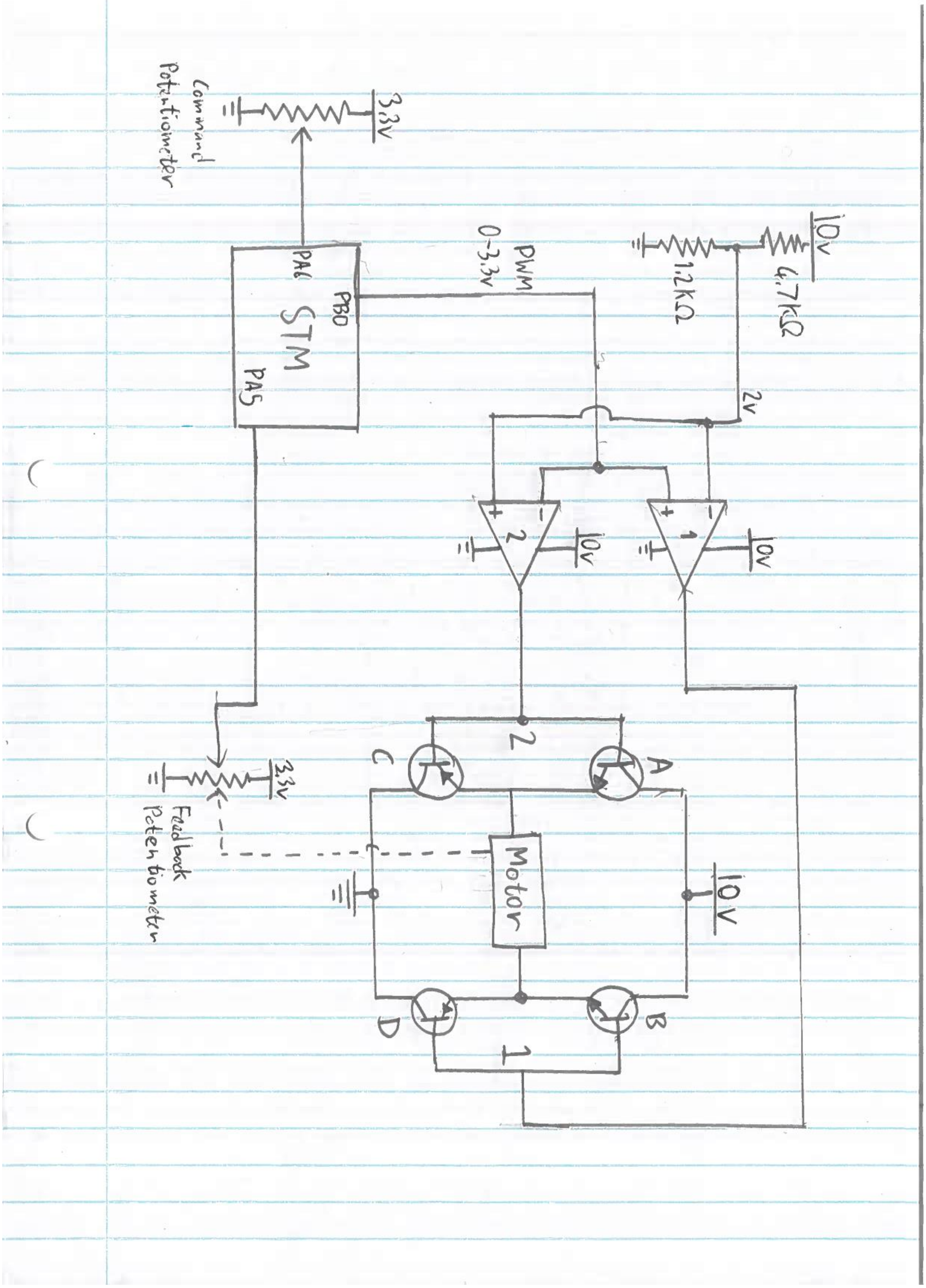
Plagiarism Declaration

By demonstrating and submitting this practical I agree that:

- I know that plagiarism is a serious form of academic dishonesty.
- I have read the document about avoiding plagiarism, am familiar with its contents and have avoided all forms of plagiarism mentioned there.
- Where I have used the words of others, I have indicated this by the use of quotation marks.
- I have referenced all quotations and other ideas borrowed from others.
- I have not and shall not allow others to plagiarise my work.
- I have not used an AI language model to generate the code or answers submitted here.

Name: Signature:

Question 1 (6 marks)



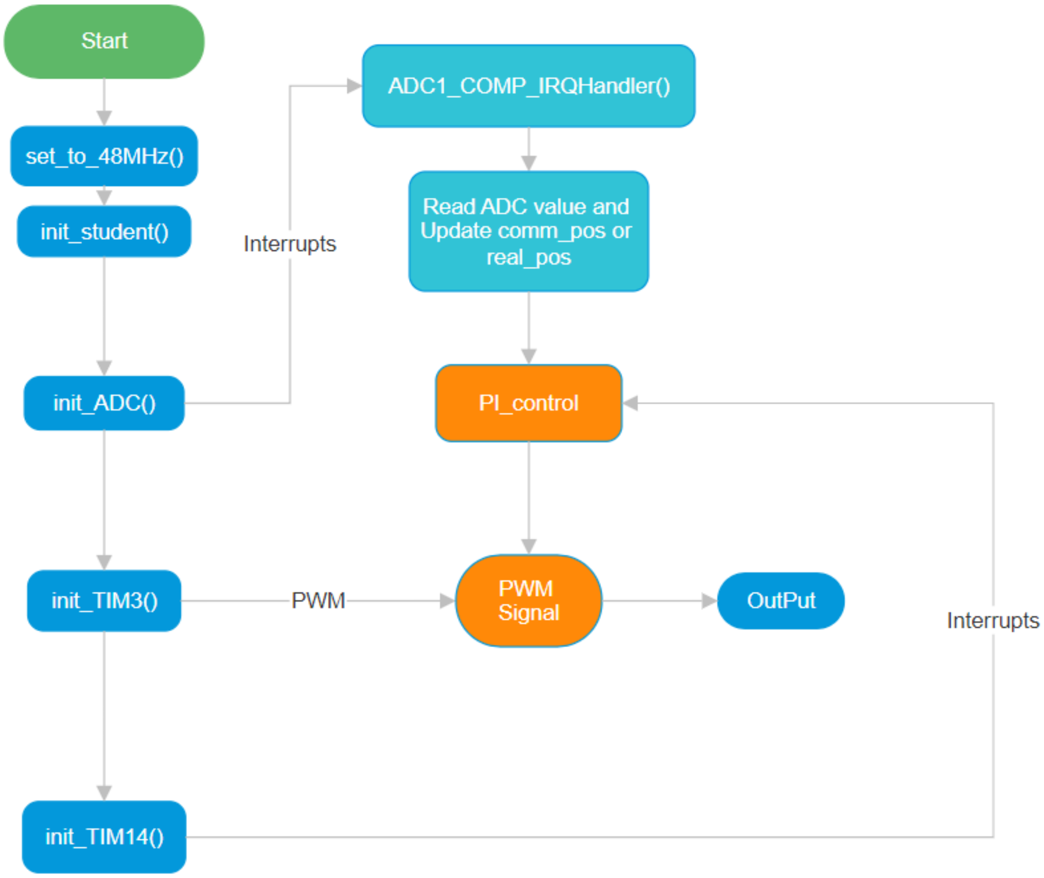
Question 2 (3 marks)

PWM	Vref (V)	Op Amp 1 (V)	Op Amp 2 (V)	DS1 (V)	DS2 (V)
High	2.3	8.3	0	7.6	0.7
Low	2.4	0	8.3	0.7	7.2

Question 3 (3 marks)

PWM	DS1 (V)	DS2 (V)	A	B	C	D	Vload (V)
High	7.6	0.7	OFF	ON	ON	OFF	3.2
Low	0.7	7.2	ON	OFF	OFF	ONN	2.9

Question 4 (14 marks)



Question 5 (6 marks)

The proportional gain (K_p) controls how strongly the system reacts to the current error between the target position and the current position. When K_p is too high, the system reacts really fast but overshoots and oscillates. When K_p is too low, the reaction is sluggish and might never actually end up at the destination accurately.

The integral gain (I) eliminates steady-state error by summing over time. This means even small errors will be eradicated over time. Excessive integral gain, however, causes overshooting and makes the system oscillate or unstable.

Sample time (T_s) controls the rate at which the PI controller updates. A low T_s (high frequency) causes the controller to update more often, thereby fastening the response of the system but also possibly noisier. A rise in T_s slows the control loop, resulting in a stable but slow-responding system.

Question 6 (10 marks)

To figure out how fast the signal can change, we look at the LM358's slew rate. It can only swing 10V at a maximum rate of 0.3 V/µs, so the change takes about 33.33 microseconds.

The PWM signal has to be slow enough that this change time is less than 1% of the total period. That means the PWM period must be at least 3333 microseconds, which corresponds to a maximum frequency of 300 Hz.

Timer Frequency = System Clock / ((PSC + 1) × (ARR + 1))

PSC = (System Clock / (Timer Frequency × (ARR + 1))) - 1

Given:

- The system clock is 48 MHz,
- We want a timer frequency of 300 Hz,
- And the auto-reload register (ARR) is set to 1023 for 10-bit resolution.

PSC = (48,000,000 / (300 × 1024)) - 1 = 156.25 - 1 = 155.25

round down to **155**.

This value ensures that the PWM signal changes slowly enough so the LM358 can keep up without signal distortion.

7 EXTRA PAGE 8 EXTRA PAGE